

Elective tracheostomy tube change

Two trained people. Child settled, at least two hours after a feed. Never alone for a first change, a difficult previous change, or an unstable child.

CHECK ALL EIGHT BEFORE YOU TOUCH THE CHILD

- ☐ Roles agreed — one changes, one holds and monitors
- ☐ New tube, **SAME size**, obturator in
- ☐ Spare tube, **HALF SIZE SMALLER**, open and within reach. A larger tube is never the fallback
- ☐ Water-soluble lubricant, sterile saline, gauze, gloves
- ☐ Suction tested; correct catheter size, depth marked
- ☐ Self-inflating bag, masks for face **and** stoma, oxygen on
- ☐ Sats probe on and alarming; capnography if ventilated
- ☐ New ties cut, blunt scissors, shoulder roll ready

SAY THIS OUT LOUD BEFORE YOU START

“If this tube does not pass, I will reposition and retry the same size once, then the half-size once, and if that fails I will stop, ventilate the face with the stoma occluded, and call for help.”

Turn over for the eight steps and the failed-change ladder. AARC CPG, *Respir Care* 2021;66(1):144–55; ATS 2025.

The change, and what to do if it fails

Nobody forces anything, at any point.

THE CHANGE — EIGHT STEPS

- 1 **Position.** Supine, shoulder roll, neck extended, head midline. Assistant steadies the head and tube.
- 2 **Pre-oxygenate and suction.** Suction to the pre-measured depth, then the mouth and nose.
- 3 **Prepare the new tube.** Obturator fully in, tip lubricated. If cuffed: test-inflate, then fully deflate.
- 4 **Release.** Assistant holds the flange, then cut the old ties. The tube is now held by hand.
- 5 **Remove.** Deflate the cuff. Withdraw along the curve of the tract in one movement, at end-inspiration.
- 6 **Insert.** Perpendicular to the neck, then rotate downward along the arc of the trachea. **No force.**
- 7 **Remove the obturator IMMEDIATELY.** It occludes the entire lumen — the commonest lethal omission.
- 8 **Secure and confirm.** Hold the flange until the ties are fastened — one finger under the tie.

IF THE NEW TUBE WILL NOT PASS

- 1 **Do not force.** Force is what makes a false passage.
- 2 **Reposition and retry the SAME size, once.** Most tubes that will not pass are simply mispositioned.
- 3 Still not passing: **half-size smaller, once.**
- 4 Two failed attempts — **STOP.** Ventilate the **face** with the stoma occluded. Use the stoma only if the upper airway is not patent. Call for help.

CONFIRM PLACEMENT THREE WAYS

Chest rise and equal breath sounds · a suction catheter passes to the marked depth and no further · capnography where available. Never confirm by how far the tube went in.

If it is not within arm’s reach, it does not exist

Check at every handover and sign below. An unchecked box is a documentation exercise, not a safety system.

EIGHT ITEMS — EVERY BED, EVERY SHIFT

- ☐ Spare tube — same size
with obturator
- ☐ Spare tube — half size smaller
with obturator
- ☐ Suction unit + catheters
correct size, pre-measured depth marked
- ☐ Self-inflating bag + masks
sized for the face AND for the stoma
- ☐ Oxygen source + tubing
enough for face and stoma at once
- ☐ Water-soluble lubricant
and sterile saline
- ☐ Spare ties and scissors
blunt-ended
- ☐ Pulse oximeter
waveform capnography where available

ALSO AT THE BEDSIDE

- ☐ Bedhead sign complete and current — upper airway status, tube make and size, date, cuffed or not, suction depth, who to call
- ☐ Both spare tubes travel with the child — to imaging, to theatre, to the ward, and home

CHECKED BY

☐ Early shift — name / time _____

☐ Late shift — name / time _____

☐ Night shift — name / time _____

THE TWO-MINUTE TEST

Ask anyone to bring you this box. Over two minutes, or one item missing, and it has failed. Fix it now, not at the next audit.

The 5 R's for safe tracheostomy care

One card, five checks. Use it at every handover.

FIVE CHECKS

R RIGHT TUBE — size increases with age. Change **every 1–2 weeks in hospital, monthly in chronic care**. Reprocess and reuse only as the manufacturer allows. Cuff only for positive-pressure ventilation or aspiration risk.

R RIGHT TAPES — secure to one-finger tightness. Change daily, clean the stoma twice a day, inspect the skin under the flange at every change.

R RIGHT HME — limits blockage by secretions. Weight-matched, under or over 10 kg. **Change at least daily, and at once if wet, soiled or blocked**. Use heated humidification in week one, above 2 L/min oxygen, or if secretions are thick.

R REGULAR SUCTION — right size, right depth, right pressure. Pre-measured depth only; never suction blindly beyond the mark.

R RESUSCITATION AWARENESS — follow the emergency algorithm. Emergency box checked every shift. Keep calm.

TWO PLACES THE ORIGINAL CARD IS OUT OF DATE

- **Catheter size.** The card says double the tube size (7 Fr for a 3.5). AARC caps the external diameter under 50% of the internal diameter — use **5 Fr**.
- **Depth.** The card says ½ cm beyond the tube tip. Use **pre-measured shallow** suction — the side holes just clear the tip, and no further.

HOW TO USE IT

Tape inside the emergency box lid; give a copy to every family at discharge. A memory aid, not a protocol — the numbers live in your unit policy.

Adapted from the RMCH “5 R's”, English and Hipson 2018 (tracheostomy.org.uk). Suction and interval: AARC CPG *Respir Care* 2022;67(2):258–71 and 2021;66(1):144–55. HME and humidification: AARC CPG *Respir Care* 2012;57(5):782–8. Reuse: manufacturer IFU.